

Automatic Evaluation of Student Math Answers Using Fine-Tuned Large Language Models

Abstract

This project investigates the use of instruction-tuned large language models (LLMs) for automatically evaluating student mathematical answers. Instead of solving mathematical problems directly, the system evaluates student submissions using a rubric-based grading scheme ranging from 0 to 4. A synthetic dataset was constructed from GSM8K-style arithmetic word problems, reference solutions, generated student submissions, rubric scores, and rationales.

Multiple model settings were evaluated, including base and fine-tuned versions of Mistral, Qwen2.5-Math, and Gemma-3-27B-IT. Fine-tuning consistently improved performance across all evaluated models. The best-performing configuration achieved approximately 90% accuracy and strong Quadratic Weighted Kappa agreement.

Error analysis showed that most remaining mistakes occurred between adjacent rubric labels, especially scores 2 and 3. Human evaluation experiments further showed that even human annotators sometimes disagreed on these borderline cases and occasionally assigned scores outside the original expected range.

1. Introduction

Automatic grading and educational feedback systems are important in online learning and intelligent tutoring environments. Many grading systems focus mainly on whether the final answer is correct, but evaluating the quality of a student's reasoning is more difficult.

Large language models have shown strong performance on many natural language processing and mathematical reasoning tasks. However, solving a problem and evaluating another person's partially correct solution are different tasks. In this project, the model is not asked to solve the problem from scratch. Instead, it is asked to evaluate a student submission against a reference solution and an explicit grading rubric.

The model receives:

- the original task
- a reference solution
- a student submission
- and a grading rubric

The model must then return:

- a score from 0 to 4
- and a short rationale explaining the assigned score

Unlike traditional mathematical reasoning benchmarks, this project evaluates the ability of LLMs to judge partially correct reasoning rather than produce solutions directly.

The main goal of this project is to measure how much fine-tuning improves rubric-based mathematical answer evaluation, and to analyze the most common remaining errors. Special attention is given to the boundary between scores 2 and 3, because this distinction requires judging whether an error is a major reasoning/setup problem or only a minor final-step mistake.

2. Related Work

GSM8K introduced a benchmark of grade-school math word problems designed to evaluate multi-step arithmetic reasoning in language models. Chain-of-thought prompting later showed that producing intermediate reasoning steps can improve large language model performance on arithmetic, commonsense, and symbolic reasoning tasks.

Recent work has also explored using LLMs as evaluators or judges for generated outputs. This project focuses specifically on rubric-based evaluation of student mathematical answers rather than general chat response evaluation.

For rationale evaluation, this project uses automatic text-similarity metrics. BERTScore measures semantic similarity using contextual embeddings, while ROUGE-L measures lexical overlap.

3. Dataset Construction

The dataset was based on GSM8K-style arithmetic word problems involving multi-step reasoning. Because GSM8K does not provide student mistakes, grading labels, or rubric-based rationales, synthetic student submissions were generated.

Each dataset entry contains:

- a task
- a reference solution
- a student submission
- an explicit rubric
- a score
- and a rationale

The grading rubric used five ordinal labels:

0 → Irrelevant answer or no useful mathematical reasoning.

1 → Uses some problem information but follows the wrong mathematical method.

2 → Partially correct reasoning, but a major conceptual, setup, logic, or calculation error prevents a correct solution.

3 → Mostly correct reasoning with a minor arithmetic or final-step mistake, or a correct/nearly correct answer with incomplete or weak reasoning.

4 → Correct reasoning with a justified solution path and correct final answer.

The most difficult distinction is between score 2 and score 3.

Dataset split:

- 70% training
- 10% validation
- 20% testing

4. Methodology

The task was formulated as rubric-based evaluation rather than mathematical solving. Each model receives the task, reference solution, student submission, and rubric. It must return a fixed score and a rationale.

The expected output format was:

Score: <0-4>

Rationale: <brief explanation>

The following model settings were evaluated:

- Base Mistral
- Fine-tuned Mistral
- Fine-tuned Qwen2.5-Math
- Base Gemma-3-27B-IT
- Fine-tuned Gemma-3-27B-IT

Fine-tuning was performed using:

- Unsloth
- LoRA
- 4-bit quantization
- BF16 precision
- SFTTrainer
- Chat-formatted training data

Main training configuration:

- Sequence length: 2048
- Batch size per device: 4
- Gradient accumulation: 4
- Effective batch size: 16
- Optimizer: adamw_8bit
- Maximum steps: 800

The Gemma fine-tuning run additionally used partial layer unfreezing with only the last transformer layer being trained together with LoRA adapters.

5. Evaluation Metrics

The following metrics were used:

- Accuracy
- Mean Absolute Error (MAE)
- Quadratic Weighted Kappa (QWK)
- BERTScore F1
- ROUGE-L
- Consistency Rate

Accuracy measures exact score prediction performance.

MAE measures average prediction distance.

QWK measures ordinal agreement while penalizing larger grading differences more heavily.

BERTScore and ROUGE-L compare generated rationales against reference rationales.

The consistency metric is heuristic-based and intended only as a lightweight indicator of rationale-score alignment.

6. Experimental Results

Fine-tuning significantly improved the performance of all evaluated base models.

Main results:

Base Mistral:

- Accuracy: 0.5857
- MAE: 0.4534
- QWK: 0.8507

Fine-tuned Mistral:

- Accuracy: 0.8633
- MAE: 0.1453
- QWK: 0.9551

Fine-tuned Qwen:

- Accuracy: 0.8894
- MAE: 0.1106
- QWK: 0.9716

Base Gemma-27B:

- Accuracy: 0.7587
- MAE: 0.2565
- QWK: 0.9278

Fine-tuned Gemma-27B:

- Accuracy: 0.9002

- MAE: 0.0998
- QWK: 0.9744

The Gemma model achieved the highest overall grading performance.

However, Gemma produced weaker rationale quality according to ROUGE-L evaluation, indicating that high rubric-score accuracy does not necessarily imply strong explanation similarity.

The Qwen and Gemma models both reduced confusion between scores 2 and 3 compared with the fine-tuned Mistral model.

7. Error Analysis

Most remaining errors occurred between adjacent rubric levels, especially scores 2 and 3. This means the models usually predicted nearby grades rather than making severe mistakes across distant labels.

Common error types included:

- arithmetic slips
- incomplete reasoning
- contradictory final statements
- answer-only submissions
- unsupported corrections
- partially correct reasoning with broken logic

Catastrophic grading failures were rare.

The main remaining difficulty was distinguishing:

- partially correct reasoning with major conceptual/setup errors
- from:
- mostly correct reasoning with only minor arithmetic or final-answer mistakes

8. Human Evaluation

A small-scale human annotation study was conducted using ambiguous grading examples. Two human annotators independently evaluated selected samples using the same rubric without seeing the original labels or the model predictions.

The primary goal of this analysis was to investigate whether the distinction between nearby rubric levels, especially scores 2 and 3, is inherently difficult even for human evaluators.

The manually reviewed samples showed that several cases were genuinely ambiguous. Some submissions contained mostly correct reasoning with only a final-step arithmetic mistake, while others contained partially correct reasoning but also introduced unsupported corrections, contradictory conclusions, or weak justifications.

Interestingly, the human annotators did not always agree with the original dataset labels. In several cases, annotators assigned scores outside the original expected boundary, including scores 0 or 1 for examples that were originally selected as ambiguous score 2/3 cases.

This observation is important because it suggests that part of the remaining model error may be caused not only by model weakness, but also by:

- ambiguity in the grading rubric
- inconsistency in synthetic labels
- and subjective interpretation of partially correct reasoning

The human evaluation therefore supports one of the central findings of this project: distinguishing between major reasoning errors and mostly correct reasoning with minor mistakes is inherently difficult, even for humans.

9. Limitations

This project has several limitations:

- The dataset is synthetic rather than collected from real students.
- Some generated labels and rationales may contain inconsistencies.
- The human evaluation subset is relatively small.
- The distinction between scores 2 and 3 is inherently ambiguous.
- Automatic rationale metrics do not fully measure logical correctness.

10. Conclusion

This project demonstrated that rubric-based mathematical answer evaluation is a feasible and promising application for instruction-tuned LLMs, especially when combined with structured prompting and task-specific fine-tuning.

Fine-tuning dramatically improved grading performance across all evaluated models. The fine-tuned Gemma-3-27B-IT model achieved the best overall performance with approximately 90% accuracy and strong ordinal agreement.

However, the remaining challenges mainly involve ambiguous reasoning boundaries rather than complete mathematical failure.

Future work should focus on:

- larger human-annotated datasets
- clearer grading rubrics
- stronger rationale evaluation metrics
- and improved dataset consistency

11. References

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